ROUND 3 / RESEARCH NOTE / 01

Can the shared propagator halve the cumulant bill?

Random ReLU MLP, width 1024, depth 16
Research completed 22 September 2026 | English report | Astra Ultra

Setup: no biases; weights N(0, 2/n); Gaussian input N(0, I). Requested cost: at most 1.2 u/source-layer or 8 u/layer for the old tier, with 2% D21 accuracy. Final raw MSE target: about 1.6e-8. Total budget: 1024 u; 1 u = 2^{31} FLOPs.

Answer

I did not establish an algorithm that meets both the requested cost and 2% slice-error targets. The shared propagator identity and the low Gaussian-chaos degree are exact, but their straightforward aggregate representations restore the pair-feature or cubic-core costs. There is also no justified universal 2 u/source-layer lower bound, and the supplied leaderboard numbers do not identify what the leaders compute.

The most consequential finding is a mismatch between the stipulated tensor and the actual code: **the two-propagator-leg source is the B3 block, not the entire third-cumulant birth.** Both the local reference implementation and V29 also retain a covariance-path B1 block. Any claimed replacement must cover both. [L1, L2]

Finding	Result
Shared squared propagator	Exact closure requires cross-column products. At r = 256, evaluating the explicit pair lift costs 32.125 u per source, versus 0.25 u for QX followed by squaring.
Hermite / Wick representation	The realized third chaos must remain. Its degree is small; its general coefficient tensor is not.
Adjoints / model reduction	Useful for testing which directions affect the output. No demonstrated small dual state or shorter exact propagation horizon.
One more cumulant order	Selected channels can cost $O(n^2)$ or a few matrix products; full augmentation and K4 are substantially more expensive. A 1.3x accuracy gap does not prove their use.

What was actually checked

I inspected the relevant source construction, slice formation, and approximation code; consulted primary research; and ran a NumPy diagnostic on the saved seed-7, layer-14, 1024-width network dump. The check script completed in 1.10 seconds. It verified the algebra and measured squared-leg norms. **No new end-to-end prediction MSE or leaderboard result was produced.**

Evidence labels in this report: **Derived** means algebra or explicit operation counting; **Checked** means executed locally; **Reported** means supplied experiments or existing logs; **Open** means an unverified route. Previously failed trace closures, hub screening, sketches and pooled low-rank fits are treated as constraints, not proposed again.

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First fix the object and the cost baseline

The birth contains more than from_dstensor

The local reference builds all-distinct covariance-path terms with `pK_111.add_factors_`, then returns their sum with `FactoredTensor.from_dstensor(K_ds[3])`. The latter has two identity legs, but the former generally does not. This is explicit at lines 400-407 and 526 of factor_k3.py. [L1]

```
B1 = Sym sum_j (3 a_j) (x) e_j (x) (w2_j a_j)
B3 = Sym sum_j m_j     (x) e_j (x) e_j
a   = diag(w1) C_off
After transport:  A = P a,     P = propagator
M   = P diag(s) + 3 A diag(e) + Z L^T
```

Here (x) denotes a tensor product and Sym is the average over permutations. In V29, A is the transported covariance leg; it is **not** the prompt's transported B3 birth matrix. Ignoring thin feedback for this display, the complete merged slice includes

```
D21 = 2 [(A*P) diag(w2)] A^T
     +  [(A*A) diag(w2)] P^T
     +  [(P*P) M^T + 2 (P*M) P^T] / 3 .
```

V29 merges these into two dense right-factor contractions and already avoids a separate dense transport of M. Its shared-basis machinery and batched hub summation must be credited before proposing any new sharing. [L2] A small explicit tensor check confirmed that B1 has nonzero all-distinct entries at birth, whereas B3 has none.

The 4-product count is a schedule, not a lower bound

With $u = 2n^3 = 2^{31}$ FLOPs, four ordinary square products cost 4 u, or about 2.04-2.05 u at the supplied Strassen setting. But the newborn P is copied from W, and the last layer needs D3 rather than the full D21. These savings already exist in V29.

```
Naive 16-layer source accounting: 4 * 120 = 480 products
Free newborn P:                   subtract 15
No final D21:                     subtract 30
Remaining illustrative total:     435 * 0.51 = 221.85 u
```

This is a count for the uncompressed main source operations, before covariance work, thin terms, basis changes and other overheads. It is not a prediction of the measured V29 bill. Dividing a whole-program cost by 120 cannot prove a per-source arithmetic lower bound.

Convention used below: unless explicitly labelled Strassen or quoted from the prompt, costs are ordinary multiply-add counts divided by $2n^3$. A rectangular (a x b)(b x c) product costs approximately abc/n^3 u. The factor 0.51 is not automatically valid for every rectangular shape or tensor contraction.

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Direction I: sharing the squared propagator

Derived. Write T = QX, where Q has n rows and r columns. The exact identity is

```
(QX)*(QX) = (Q*Q)(X*X)
  + 2 sum_{a<b} (q_a*q_b) (x_a*x_b) .
Pair feature dimension: s = r(r+1)/2 .
```

The final products are outer products of Q columns and X rows. The missing terms are cross-column features, not a correction that follows from Q*Q. For example, rows (1,1) and (1,-1) have identical squares; multiplying each by $(1,1)^T$ and then squaring gives 4 and 0. Thus no function of Q*Q and X alone can recover the answer for all signed Q.

(a) State and (b) update cost

The explicit lift stores Phi(Q), an n x s pair-feature matrix, and Psi(X_b), an s x n matrix per birth. A change Q ← MQ costs $2n^2r$ FLOPs, or r/n u, before rebuilding the features in O(ns) operations. Multiplying the two feature matrices costs $2n^2s$ FLOPs, or s/n u per source. Keeping signed Q and X themselves uses fewer numbers, but then ordinary QX is the natural exact evaluation.

r	s	Phi, FP32	Lift eval	Direct QX
224	25,200	98.44 MiB	24.609 u	0.21875 u
256	32,896	128.50 MiB	32.125 u	0.25000 u
384	73,920	288.75 MiB	72.188 u	0.37500 u
1024	524,800	2050 MiB	512.500 u	1.00000 u

These lift prices exclude feature construction and subsequent multiplication by the third leg. If the third leg also lies in Q, pooling its coefficients replaces all per-birth Psi tables by a third-order core. That is the pooled-core route already measured in item 7, rather than a new n^2 recurrence.

(c) Why the odd signal survives, and (d) falsifier

Keeping the full pair lift and the signed third leg is exact, so the full cubic slice changes sign when the current W changes sign. Keeping only diagonal pair features removes realized cross terms even though it can still produce an odd answer. Oddness is necessary, not sufficient.

Run the accompanying script on query_dump_L14.npz. It orthonormalizes the stored basis and compares (P*P) with (Q*Q)(FP*FP). **Checked:** relative errors were 79.22% and 80.85% for old slots 0 and 9. The exact pair identity on a small matrix agreed to 2.93e-16. These are matrix errors, not new D21 or final-mean errors; they reject the proposed square-only algebra, not every possible compression.

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Direction II: low-order chaos is not a small state

Derived. Freeze all incoming factors and previous-layer Wick coefficients. Absorb the diagonal Wick map into a symmetric input tensor K, and write the independent current weight rows as $w_i \sim N(0, \sigma^2 I)$, $\sigma^2 = 2/n$. Then the transported slice is $D_{ik} = K(w_i, w_i, w_k)$. Let $\tau_c = \sum_a K_{aac}$.

```
For i != k:
D_ik = sigma^2 tau.w_k
     + sum_abc K_abc (w_ia w_ib - sigma^2 delta_ab) w_kc

For i = k:
D_ii = 3 sigma^2 tau.w_i + K : :w_i^(x3): .
```

The second term in each expression is third Gaussian chaos (degree two in one row and degree one in the other when i differs from k). It is not made cheaper by naming it a Wick polynomial. Its exact coefficient object is K again. The same observation applies to the degree-two chaos in an individual squared propagator.

(a) State and (b) price

A general symmetric K has $n(n+1)(n+2)/6$ independent entries; an unpacked n^3 FP32 array is 4 GiB at $n = 1024$. In a common r-dimensional basis, use Q plus a cubic core, $O(nr + r^3)$ numbers. A general dense basis rotation costs about $6r^4$ FLOPs ($3r^4$ multiply-add pairs); a direct slice evaluation costs $O(nr^3 + n^2r)$. The optimistic symmetric pricing in the prompt is assessed on page 6. Alternatively keep the source factors: that returns to the original sourcewise computation.

(c) The fluctuation is quantitatively present

```
i != k, conditional on K:
E[(sigma^2 tau.w_k)^2] = sigma^6 ||tau||^2
E[(third-chaos term)^2] = 2 sigma^6 ||K||_F^2
The two components are orthogonal in Gaussian L2.
```

The covariance/Gram conditional mean loses the odd contribution, as in the supplied experiments. Retaining only the first-chaos term is a trace closure already tested and rejected. Full third-chaos coefficients retain the realized signal; averaging them away does not.

The stated 1/sqrt(n) fluctuation scale needs a norm qualifier. For a Gaussian entry g, $E[(g^2 - Eg^2)^2] = 2(Eg^2)^2$ and $E[g^4] = 3(Eg^2)^2$. Thus centered squares carry approximately sqrt(2/3) = 81.65% of Frobenius norm. A 1/sqrt(n) ratio can instead describe an operator-norm comparison with a coherent rank-one mean. **Checked on the real dump:** the Frobenius ratios were 81.584% and 81.604%.

(d) Falsifier

The included check verifies the mixed-row and same-row Wick identities to about 2.5e-14 absolute error. For any claimed reduced coefficient state, compare its realized third-chaos contraction with the factorized reference, after subtracting the exact first-chaos part; reject at relative RMS error above 2%. Reporting total-slice error alone can hide a failed fluctuation approximation.

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Direction III: the adjoint moves the same difficulty

Derived for a frozen linear-response chain. Let L_l apply M_l to each of three tensor legs, and let S extract the required slice. For a scalar output probe v, the Gaussian-chain adjoint supplies matrices G_l such that the first-order correction is a sum of slice contractions.

```
K_l = L_l K_(l-1) + B_l
J_v = sum_l <G_l, S K_l>
Lambda_l = S* G_l + L_(l+1)* Lambda_(l+1)
J_v = sum_b <Lambda_b, B_b>    (zero initial K)

S*G = Sym sum_i e_i (x) e_i (x) G[i,:] .
```

This is the useful exact identity: all future queries can be accumulated into a dual state. However, each layer injects another hub tensor into Lambda, and backward transport sends all three legs through M^T. It therefore has the same sum-of-births structure in reverse. The injection already has up to n hub columns, even for one scalar final-output probe.

(a) State, (b) cost, and (c) parity

An unpacked dense scalar dual state uses n^3 numbers; three dense mode rotations cost about $6n^4$ FLOPs = 3n u = 3072 u/layer at width 1024. An r^3 core has the pooled-core costs. A source-factor dual has growing hub rank and sourcewise transports. Obtaining G_l through a Gaussian mean/covariance adjoint may cost only $O(Ln^3)$, but this does not include transporting Lambda.

One scalar adjoint returns v^Tmu, whereas the task requests all 1024 coordinates of mu. Batching output probes requires another justified reduction or additional cost. Reversing the exact contractions retains the realized signs. Truncating them based only on even statistics does not.

The remaining opening: output-aware compression

Balanced truncation and CoBRAS use both state excitation and output sensitivity, rather than state energy alone. [R5, R6] This is a legitimate way to test for a smaller effective state. It is **not yet a priced solution here**: one must construct an inexpensive basis, propagate it across the changing random layers, and bound nonlinear feedback error. The measured two-mode response result does not prove a general balanced-rank lower bound, but supplies no evidence of the very small rank required.

(d) A bounded falsifier, not a new expensive benchmark

For a proposed scalar dual implementation, first verify the displayed duality identity against forward factor contractions on a frozen trajectory. Use 32 fixed Rademacher slice probes and compare the vector of scalar answers in RMS, not each near-zero scalar separately. A 2% failure rejects the dual representation. Time-box the probe batch at 30 minutes and count all factor transports and basis construction. A pass is only a screening test, not certification for the full output vector.

For a claimed fixed short horizon, the supplied item-5 ablations already reject simple deletion of early source-query effects; they need not be rerun. Checkpointing or meeting forward and backward computations halfway changes storage and scheduling, but does not by itself remove these contractions.

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Direction IV: what closure literature actually offers

Representation	State / update	Application to this task
CE3; CE2.5; EDQNM [R2]	CE3 retains third-order statistics. CE2.5 replaces their evolution by an approximate diagnostic equilibrium.	A damping parameter or quasi-steady closure does not reconstruct this network's missing odd realization. The supplied closure and memory tests are adverse evidence.
Mori-Zwanzig [R3]	Resolved variables plus a memory kernel and an unresolved-force contribution. Exact projection is generally non-Markovian.	The unresolved force carries the information a slice-only state lost. Replacing it by its ensemble mean or independent noise cannot recover a fixed network's phase.
TT / dynamical low rank [R4, R7]	A third-order TT of mode size r and ranks k1,k2 stores r k1 + r k1 k2 + r k2 entries.	At k1,k2 near r, storage is cubic again. These methods organize approximation; they do not ensure small unfolding ranks for this tensor.

(a) A pooled core and (b) its favorable cost ledger

Use Q (n x r) and a symmetric core C (r x r x r). Grant the prompt's optimistic costs: slice $n\,r^3/2$, join $n\,r^3/2$, and rotation $3r^4$ FLOPs. Dividing by $2n^3$ gives the following. Standard dense kernels can cost more; this is deliberately a favorable ledger, not a benchmark of a packed symmetric kernel.

r	Core FP32	Slice	Join	Rotation	Total
160	15.6 MiB	0.977 u	0.977 u	0.916 u	2.869 u
224	42.9 MiB	2.680 u	2.680 u	3.517 u	8.876 u
250	59.6 MiB	3.725 u	3.725 u	5.457 u	12.908 u
256	64.0 MiB	4.000 u	4.000 u	6.000 u	14.000 u
384	216 MiB	13.500 u	13.500 u	30.375 u	57.375 u

This table assumes one join and one basis rotation per layer and excludes forming or transporting Q, QR, final output projection, and thin channels. Avoiding a rotation during source-free transport is possible by moving Q with M; handling the new old-source join without growth or a coordinate change is then the unresolved cost. Amortized joins would require a separately priced cohort schedule.

(c) Signal and (d) falsifier

A full core retains the odd signal; a projection keeps the actual retained coefficients. At the reported rank near 250, this schedule costs 12.91 u before extras. A faster packed or rectangular kernel could change the ledger: multiplying all entries by 0.51 without implementing and metering the relevant shapes is not justified. This is a rejected schedule, not a lower bound on all pooled methods. Any TT claim must include its middle core and conversion costs.

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A real sub-2-u special case, with an explicit limit

It is useful to exhibit a case that defeats the alleged arithmetic floor, without claiming it fits these networks. Suppose the stipulated B3 birth matrix is **exactly diagonal**, A_b = diag(d_b). Then its three-leg symmetrization becomes a sum of weighted cubes:

```
Atilde_b = T_b diag(d_b)
D21_b = [(T_b*T_b) diag(d_b)] T_b^T .
State: T_b (n x n), d_b (n).
Update: T_b <- M T_b.
Cost: one transport + one slice product = 2 u ordinary
      approximately 1.02 u at the supplied Strassen setting.
```

This state keeps every hub and every birth. Its slice is odd, realized, and potentially full rank. Therefore neither oddness nor full rank proves a 2 u/source-layer lower bound.

Why it is not a demonstrated approximation for this project

Writing A_b as diagonal plus entrywise O(1/sqrt(n)) noise does not make that noise small in the required norm. A dense matrix with n^2 entries of that size can have O(n) squared Frobenius norm and O(1) operator norm, comparable to an O(1) diagonal. Moreover, the actual B1 covariance-path source still needs to be retained.

A rank-k residual A_b = diag(d_b) + U_b V_b^T is also not a free fix. Carry U = T_b U_b and static V_b. The exact additional slice is

```
Delta D21 = [(T*T)V] U^T / 3
          + 2 [(T * (U V^T)) T^T] / 3 .
```

The first term costs about 2k/n u using two thin products. The second still requires a dense product after forming U V^T, or k weighted Gram products after expanding it. For the straightforward thin-then-dense implementation, the extra cost is one square product plus O(k/n) u. With transport and the diagonal part, even this special low-rank-residual construction has about three square products, around 1.53 u, before thin work and any birth compression. This is an algorithm count, not a lower bound.

Falsifier and status

If considering the diagonal-only case, use a stored **B3 birth** matrix, not the file's covariance A leg: carry its diagonal and residual separately through the same frozen propagators, then measure their contracted D21 errors. A residual above 2% rejects the approximation. This needs two dense slice contractions per saved frame, rather than Monte Carlo truth regeneration. Existing evidence gives no basis to expect a pass, and this report does not recommend repeating a diagonal/trace closure already ruled out.

Result: there is an exact cheap special case, but no demonstrated birth-matrix approximation that turns V29 into that case. It cannot explain the leaders without further evidence.

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The leaderboard gap is not an algorithm identifier

The two cost targets are different

Reported V29 total = 260 u, of which the old tier costs 107 u. Everything else therefore costs 153 u. The old tier spans approximately ten relevant deep layers in the stated schedule. Reducing it to 8 u/layer would cost roughly 80 u and leave the whole program near **233 u**, not 133-165 u. Exact totals depend on boundary layers and what is included in the tier counter.

Whole-program target	Required saving	What this implies if other costs stay fixed
165 u	95 u	Old tier would have to fall from 107 u to 12 u: about 89% cheaper.
133 u	127 u	Deleting all 107 u would still leave a 20 u gap elsewhere.
8 u per old layer	About 27 u in a ten-layer illustration	A useful intermediate target, but insufficient to explain the full leaderboard gap.

An accuracy ladder does not prove a higher cumulant order

A constant-factor MSE improvement at a different width, or on a different set of networks, cannot identify the leaders' internal state. The improvement might involve different terms, their approximation, numerical details, or cancellation of baseline bias. These are possibilities, not claims about a particular submission. No leader source code was inspected in this research.

Separate perturbation error from error against truth

```
e = baseline prediction - truth;  delta = new - baseline
MSE_new = MSE_base + mean(delta^2) + 2 mean(e*delta) .
```

The measured rule 4.2e-6 epsilon^2 is useful for the perturbations on which it was fitted; it is not a universal lower bound for all representations. At epsilon = 2%, its quoted extra term is 1.68e-9. Adding that to 2.13e-8 gives 2.298e-8 under a zero-cross-term assumption. Thus reaching 1.6e-8 requires a separate accuracy improvement or favorable measured error correlation, even if the compression is cheap.

If the objective were only to stay below 1.3 times the baseline MSE, the same additive model would allow epsilon near 3.90%. That weaker criterion is not the stated 1.6e-8 goal. For worst-case protection, the cross term cannot simply be omitted.

What could remain possible despite birth ablations?

Keeping output-relevant directions from *every* birth could be compatible with item 5, whereas deleting whole births is not. Output-aware projection also differs from hub screening. But neither an affordable construction nor a successful 2% validation is established here. **I cannot name a demonstrated approximation that halves this bill while losing less than 1.3x MSE.** The correct conclusion is an unresolved algorithmic gap, not that the leaders must be dropping sources.

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Pricing the next accuracy rung

1. Selected fourth-order content: cheap but already partial

V29 already uses a regenerated fourth-order matrix core G = diag(g) + lambda C_off. Its transported diagonal follows from the existing covariance transport: [L2, L3]

```
diag(W G W^T) = (W*W) g
  + lambda [diag(C_pre) - (W*W) var] .
```

This use-side quantity is $O(n^2)$. Restoring the omitted off-diagonal spill for *this assumed G* requires one weighted Gram product, W diag(g - lambda var) W^T, costing about 1 u ordinary or 0.51 u in the supplied square-product model per layer. Across 15 transitions this is about 7.65 u before any additional use/feed operations. Existing project evidence reports that dense spill transport did not improve the measured mean error; it is not a new recommended knob. [L3]

2. Full augmented K3: not just more pointwise terms

In this checkout, factor_k3.py and factored_keeps_term explicitly omit non-hypertree all-distinct diagrams and products needing the all-distinct third-cumulant block. The code names covariance triangles and kappa3-times-edge terms as examples. Thus *factored augment* and *dense augment* are distinct targets. [L1, L4]

For illustration, suppressing Wick scalars and repeated-index adjustments, a covariance-triangle tensor has K_abc = C_ab C_ac C_bc. Its transformed query K(w_i,w_i,w_k) can be evaluated by first forming

```
q_i,c = diag[C diag(w_i) C diag(w_i) C]_c
D_ik  = sum_c q_i,c w_k,c .
```

For each i, a direct implementation uses one dense matrix product plus $O(n^2)$ reductions; then assemble D with another product. That is approximately n + 1 u ordinary for a full slice, or roughly 522.75 u if every relevant square product achieved 0.51 u. This is a concrete expensive implementation, not a complexity lower bound. Keeping a general n^3 triangle/core state and propagating it is also expensive. The identity does not supply a few-n^3 all-layer replacement.

3. Full K4: the measured implementation is far outside the target

The saved synth128x16 log reports 516 u = 0.504 of the fixed 2^41 budget, confirming that u here means 2^31 FLOPs. [L5] If its observed n^4-class cost dominated at width 1024, the scaling estimate would be 516 x 8^4 = **2,113,536 u**. This is an extrapolation of that implementation, not a theorem excluding every K4 approximation. A dense n^4 FP32 tensor alone would occupy 4 TiB at width 1024; factorization reduces storage but does not make its current contractions affordable.

Price conclusion: cheap selected channels exist, and several are already in V29. No verified cheap implementation of the full omitted augmentation or full K4 rung was found. The reference paper provides the cumulant, Hermite and factorization framework; its fixed-depth asymptotics are not an n = 1024, depth-16 cost certificate. [R1]

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Reproducible checks and rejection tests

Checks executed for this report

Check	Result	Interpretation
Full pair-feature identity	2.93e-16 relative error	Exact identity verified on small random matrices.
PPA slice formula	1.45e-16 relative error	The prompt's stipulated B3 formula is correct.
Wick split, two row cases	2.49e-14 / 1.95e-14 absolute error	First and third chaos add back to the cubic query.
Birth all-distinct support	B1 norm 30.525; B3 norm 0	A small constructed example confirms the code-level structural distinction.
L14 old slot 0 / slot 9	79.22% / 80.85% square-only error	Dropping cross-column pair features fails on the saved network.
Same two saved legs	81.584% / 81.604% centered-square norm ratio	Frobenius fluctuation is substantial; this is not a mean-prediction test.

Data: query_dump_L14.npz, generated with seed 7, width 1024, depth 16; ten old sources and a 384-column basis. Leg reconstruction errors were 1.11e-6 and 1.96e-6. The diagnostic imports no estimator modules, uses allow_pickle=False and one numerical-library thread. Elapsed time: 1.10 seconds, excluding startup.

Re-run the delivered diagnostic

```
python round3_checks.py --root "PROJECT" --out results.json
```

Replace PROJECT with the full project directory. NumPy is the only nonstandard dependency. The JSON records results, cost calculations and code hashes. This offline diagnostic is not billed through the competition metering layer.

Screen future proposals before a full accuracy run

First 5 minutes: verify the B1/B3 target, units, memory footprint, exact square identity and full operation ledger. Reject a proposal that silently replaces a Khatri-Rao contraction by n^2r work or excludes joins/rotations from the bill.

Next 20 minutes: test the candidate on the saved L10 and L14 states, keeping its claimed signed inputs. Report separately: centered-square or third-chaos error; B1 and B3 slice errors where available; and the aggregated D21 error. Current query dumps contain A, P and w2, not every B3 or feedback field, so a full-D21 claim requires extending the dump rather than relabelling these checks.

Next 30 minutes: for an implemented dual or reduced-state candidate, compare 32 fixed scalar probes against exact factor contractions, with a 2% pooled RMS rejection threshold and a time cap. A passing probe batch permits a later metered, paired-truth run; it does not certify 2% error for every layer. A universal runtime guarantee cannot be made for an unspecified candidate or machine.

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Sources, provenance and scope

Primary research consulted. Links were accessed during this research on 21-22 September 2026. Mathematical identities and numerical cost tables in this note were derived here; the literature supplies context and possible representation families.

[R1] Wu et al. (2026), Estimating the expected output of wide random MLPs more efficiently than sampling. Especially sections 4.3 and S.4.3. Open primary source.

[R2] Marston, Qi and Tobias, Direct Statistical Simulation of a Jet. CE3 and CE2.5 closure assumptions. Open primary source.

[R3] Darve, Solomon and Kia (2009), Computing Generalized Langevin Equations and Generalized Fokker-Planck Equations. Includes discrete-map projection. Open primary source.

[R4] Oseledets (2011), Tensor-Train Decomposition. SIAM J. Sci. Comput. 33(5), 2295-2317. Open primary source.

[R5] Ma, Rowley and Tadmor, Snapshot-based Balanced Truncation for Linear Time-periodic Systems. Open primary source.

[R6] Otto, Padovan and Rowley, Model Reduction for Nonlinear Systems by Balanced Truncation of State and Gradient Covariance, version 4. Open primary source.

[R7] Koch and Lubich (2007), Dynamical Low-Rank Approximation. SIAM J. Matrix Anal. Appl. 29(2), 434-454. Open primary source.

Local source map

All paths below are relative to:
D:\Machine Learning 1 Year Challenge\Kaggle\ARC White-Box Estimation Challenge 2026

[L1] mlp_cumulant_propagation/src/mlp_kprop/factor_k3.py: 253 (from_dstensor), 400-407 (path factors), 526 (sum of source families); augment caveat at the nonlinear function header.

[L2] submission_v29m8f5/estimator.py: 1270 (covariance birth leg), 1433 (M decomposition), 1511 (_hub2), 1521-1538 (_dslices and final-layer trim).

[L3] whest-p2-cumulant-k3/docs/derivation_code_map.md, sections 2.3-2.4; docs/claims_evidence.md, entries B1-B3 and C1-C4. These include prior measured claims, not reruns in this report.

[L4] mlp_cumulant_propagation/src/mlp_kprop/kprop_harmonic.py: 86, factored_keeps_term and its documented exclusions.

[L5] kaggle_out_kprop5/whest-kprop-reference.log: 60, synth128x16-0 K4-simple-factored result, raw FLOPs 516 u (0.504 B).

[L6] probe_query.py, probe_query_analyze.py, probe_agg.py, query_dump_L14.npz; supplied ROUND3_prompt_propagator_structure.md for the twelve experimental constraints.

What remains unproved

There is no successful new estimator in this report, no new final-mean MSE, no complexity lower bound excluding a faster algorithm, and no evidence identifying the leaders' implementation. The concrete contributions are the source-family audit, exact expansion and adjoint identities, corrected cost comparisons, local numerical checks, and a reproducible rejection protocol. The research does not justify claiming that the requested bound has been beaten.